

established all of this in peer-reviewed studies published in journals such as *Nature*, *Cell*, and *Science*.

And in no small measure, because sirtuins do all of this based on a rather simple program—the wondrous gene B in the survival circuit—they’re turning out to be more amenable to manipulation than many other longevity genes. They are, it would appear, one of the first dominos in the magnificent Rube Goldberg machine of life, the key to understanding how our genetic material protects itself during times of adversity, allowing life to persist and thrive for billions of years.

Sirtuins aren’t the only longevity genes. Two other very well studied sets of genes perform similar roles, which also have been proven to be manipulable in ways that can offer longer and healthier lives.

One of these is called target of rapamycin, or TOR, a complex of proteins that regulates growth and metabolism. Like sirtuins, scientists have found TOR—called mTOR in mammals—in every organism in which they’ve looked for it. Like that of sirtuins, mTOR activity is exquisitely regulated by nutrients. And like the sirtuins, mTOR can signal cells in stress to hunker down and improve survival by boosting such activities as DNA repair, reducing inflammation caused by senescent cells, and, perhaps its most important function, digesting old proteins.²⁷

When all is well and fine, TOR is a master driver of cell growth. It senses the amount of amino acids that is available and dictates how much protein is created in response. When it is inhibited, though, it forces cells to hunker down, dividing less and reusing old cellular components to maintain energy and extend survival—sort of like going to the junkyard to find parts with which to fix up an old car rather than buying a new one, a process called autophagy. When our ancestors were unsuccessful in bringing down a woolly mammoth and had to survive on meager rations of protein, it was the shutting down of mTOR that permitted them to survive.

The other pathway is a metabolic control enzyme known as AMPK, which evolved to respond to low energy levels. It has also been highly conserved among species and, as with sirtuins and TOR, we have learned a lot about how to control it.

These defense systems are all activated in response to biological stress. Clearly, some stresses are simply too great to overcome—step on a snail, and its days are over. Acute trauma and uncontrollable infections will kill an organism without *aging* that organism. Sometimes the stress inside a cell, such as a multitude of DNA breaks, is too much to handle. Even if the cell is able to repair the breaks in the short term without leaving mutations, there is information loss at the epigenetic level.

Here's the important point: there are plenty of stressors that will activate longevity genes without damaging the cell, including certain types of exercise, intermittent fasting, low-protein diets, and exposure to hot and cold temperatures (I discuss this in chapter 4). That's called hormesis.²⁸ Hormesis is generally good for organisms, especially when it can be induced without causing any lasting damage. When hormesis happens, all is well. And, in fact, all is *better* than well, because the little bit of stress that occurs when the genes are activated prompts the rest of the system to hunker down, to conserve, to survive a little longer. That's the start of longevity.

Complementing these approaches are hormesis-mimicking molecules. Drugs in development and at least two drugs on the market can turn on the body's defenses without creating *any* damage. It's like making a prank call to the Pentagon. The troops and the Army Corps of Engineers are sent out, but there's no war. In this way, we can mimic the benefits of exercise and intermittent fasting with a single pill (I discuss this in chapter 5).

Our ability to control all of these genetic pathways will fundamentally transform medicine and the shape of our everyday lives. Indeed, it will change the way we define our species.

And yes, I realize how that sounds. So let me explain why.

TWO

THE DEMENTED PIANIST

ON APRIL 15, 2003, NEWSPAPERS, television programs, and websites around the world carried the story: the mapping of the human genome was complete.

There was just one pesky problem: it really wasn't. There were, in fact, huge gaps in the sequence.

This wasn't a case of the mainstream news media blowing things out of proportion. Highly respected scientific journals such as *Science* and *Nature* told pretty much the same story. It also wasn't a case of scientists overstating their work. The truth is simply that, at the time, most researchers involved in the thirteen-year, \$1 billion project agreed that we'd come as close as we possibly could—given the technology of the time—to identifying each of the 3 billion base pairs in our DNA.

The parts of the genome that were missing, generally overlapping sections of repetitive nucleotides, were just not considered important. These were areas of the code of life that were once derided as “junk DNA” and that are now a little better respected but still generally disregarded as “noncoding.” From the perspective of many of the best minds in science at the time, those regions were little more than the ghosts of genomes past, mostly remnants of dead hitchhiking viruses that had integrated into the genome hundreds of thousands of years ago. The stuff that makes us who we are, it was thought, had largely

been identified, and we had what we needed to propel forward our understanding of what makes us human.

Yet by some estimates, that genetic dark matter accounts for as much as 69 percent of the total genome,¹ and even within the regions generally regarded as “coding,” some scientists believe, up to 10 percent has yet to be decoded, including regions that impact aging.²

In the relatively short time that has come and gone since 2003, we have come to find out that within the famous double helix, there were sequences that were not just unmapped but essential to our lives. Indeed, many thousands of sequences had gone undetected because the original algorithms to detect genes were written to disregard any gene less than 300 base pairs long. In fact, genes can be as short as 21 base pairs, and today we’re discovering hundreds of them all over the genome.

These genes tell our cells to create specific proteins, and these proteins are the building blocks of the processes and traits that constitute human biology and lived experiences. And as we get closer to identifying a complete sequence of our DNA, we’ve come closer to having a “map” of the genes that control so much of our existence.

Even once we have a complete code, though, there’s something we still won’t be able to find.

We won’t be able to find an aging gene.

We have found genes that impact the *symptoms* of aging. We’ve found longevity genes that control the body’s defenses against aging and thus offer a path to slowing aging through natural, pharmaceutical, and technological interventions. But unlike the oncogenes that were discovered in the 1970s and that have given us a good target for going to battle against cancer, we haven’t identified a singular gene that causes aging. And we won’t.

Because our genes did not evolve to *cause* aging.

YEAST OF EDEN

My journey toward formulating the Information Theory of Aging was a long one. And in no small part, it can be traced to the work of a scientist who toiled without fame but whose work helped set the stage for a lot of the longevity research being done around the world today.

His name was Robert Mortimer, and if there was one adjective that seemed to come up more than any other about him after he passed away, it was “kind.”

“Visionary” was another. “Brilliant,” “inquisitive,” and “hardworking,” too. But I’ve long been inspired by the example Mortimer set for his fellow scientists. Mortimer, who died in 2007, had played a tremendously important role in elevating *Saccharomyces cerevisiae* from a seemingly lowly, single-celled yeast with a sweet tooth (its name means “sugar-loving”) to its rightful place as one of the world’s most important research organisms.

Mortimer collected thousands of mutant yeast strains in his lab, many of which had been developed right there at the University of California, Berkeley. He could have paid for his research, and then some, by charging the thousands of scientists he supplied through the university’s Yeast Genetic Stock Center. But anyone, from impecunious undergraduates to tenured professors at the world’s best-funded research institutions, could browse the center’s catalog, request any strain, and have it promptly delivered for the cost of postage.³

And because he made it so easy and so inexpensive, yeast research bloomed.

When Mortimer began working on *S. cerevisiae* alongside fellow biologist John Johnston⁴ in the 1950s, hardly anyone was interested in yeast. To most, it didn’t seem we could learn much about our complex selves by studying a tiny fungus. It was a struggle to convince the scientific community that yeast could be useful for something more than baking bread, brewing beer, and vinting wine.

What Mortimer and Johnston recognized, and what many others began to realize in the years to come, was that those tiny yeast cells are not so different from ourselves. For their size, their genetic and biochemical makeup is extraordinarily complex, making them an exceptionally good model for understanding the biological processes that sustain life and control lifespans in

large complex organisms such as ourselves. If you are skeptical that a yeast cell can tell us anything about cancer, Alzheimer's disease, rare diseases, or aging, consider that there have been five Nobel Prizes in Physiology or Medicine awarded for genetic studies in yeast, including the 2009 prize for discovering how cells counteract telomere shortening, one of the hallmarks of aging.⁵

The work Mortimer and Johnston did—and, in particular, a seminal paper in 1959 that demonstrated that mother and daughter yeast cells can have vastly different lifespans—would set the stage for a world-shattering change in the way we view the limits of life. And by the time of Mortimer's death in 2007, there were some 10,000 researchers studying yeast around the globe.

Yes, humans are separated from yeast by a billion years of evolution, but we still have a lot in common. *S. cerevisiae* shares some 70 percent of our genes. And what it does with those genes isn't so different from what we do with them. Like a whole lot of humans, yeast cells are almost always trying to do one of two things: either they're trying to eat, or they're trying to reproduce. They're hungry or they're horny. As they age, much like humans, they slow down and grow larger, rounder, and less fertile. But whereas humans go through this process over the course of many decades, yeast cells experience it in a week. That makes them a pretty good place to start in the quest to understand aging.

Indeed, the potential for a humble yeast to tell us so much about ourselves—and do so quite quickly relative to other research organisms—was a big part of the reason I decided to begin my career by studying *S. cerevisiae*. They also smell like fresh bread.

I met Mortimer in Vienna in 1992, when I was in my early 20s and attending the International Yeast Conference—yes, there is such a thing—with my two PhD supervisors, Professor Ian Dawes, a rule-avoiding Australian from the University of New South Wales,⁶ and Professor Richard Dickinson, a rule-abiding Briton from the University of Cardiff, Wales.

Mortimer was in Vienna to discuss a momentous scientific endeavor: the sequencing of the yeast genome. I was there to be inspired. And I was.⁷ If I'd harbored any doubts about my decision to dedicate the opening years of my scientific career to a single-celled fungus, they all went away when I came face to

face with people who were building great knowledge in a field that had hardly existed a few decades before.

It was shortly after that conference that one of the world's top scientists in the yeast field, Leonard Guarente of the Massachusetts Institute of Technology, came to Sydney on holiday to visit Ian Dawes. Guarente and I ended up at a dinner together, and I made sure I was sitting opposite him.

I was then a graduate student using yeast to understand an inherited condition called maple syrup urine disease. As you might imagine from its name, the disease is not something most polite people discuss over dinner. Guarente, though, engaged me in a scientific discussion with a curiosity and enthusiasm that was nothing short of enchanting. The conversation soon turned to his latest project—he had begun studying aging in yeast the past few months—work that had its roots in the workable genetic map that Mortimer had completed in the mid-1970s.

That was it. I had a passion for understanding aging, and I knew something about wrangling a yeast cell with a microscope and micromanipulator. Those were essential skills needed to figure out why yeast age. That night, Guarente and I agreed on one thing: if we couldn't solve the problem of aging in yeast, we had no chance in humans.

I didn't just *want* to work with him. I *had* to work with him.

Dawes wrote him to tell him that I was keen to join his lab and I was “skilled at the bench.”

“It would be a pleasure to work with David,” he replied a few weeks later, the same way he probably did to so many other enthusiastic applicants. “But he's got to come with his own funding.” Later I learned he had been excited only because he'd thought I was the *other* student he'd met at dinner.

I had a foot in the door, but my chances were slim. At the time, foreigners weren't considered for prestigious postdoctoral awards in the United States, but I insisted I be interviewed and paid for a flight to Boston myself. I was interviewed by a giant in the stem cell field, Douglas Melton, for a Helen Hay Whitney Foundation Fellowship, which has been providing research support to postdoctoral biomedical students since 1947. After waiting in line outside his office with the other four candidates, I had my chance. This was my moment. I

don't remember being nervous. I figured I probably wouldn't get the award anyway. So I went for it.

I told Melton about my lifelong quest to understand aging and find “life-giving genes,” then sketched out on his whiteboard how the genes work and what I'd be doing for the next three years if I got the money. To show my gratitude, I gave him a bottle of red wine that I'd brought from Australia.

Afterward, two things became clear. One, don't bring wine to an interview because it can be seen as a bribe. And two, Melton must have liked what I said and how I said it, because I flew home, got the fellowship, and then got onto a plane back to Boston. It was, without a doubt, the most life-changing meeting of my life.⁸

At the time of my arrival, in 1995, I had expected to build our understanding of aging by studying Werner syndrome, a terrible disease that occurs in less than 1 in 100,000 live births, with symptoms that include a loss of body strength, wrinkles, gray hair, hair loss, cataracts, osteoporosis, heart problems, and many other telltale signs of aging—not among folks in their 70s and 80s but rather among people in their 30s and 40s. Life expectancy for someone with Werner is 46 years.

Within two weeks of my arrival in the United States, though, a research team at the University of Washington, headed by the wise and supportive grandfather of aging research, George Martin, announced that they had found the gene that, when mutated, causes Werner syndrome.⁹ It was deflating at the time to have been “scooped,” but the discovery allowed me to take a bigger first step toward my ultimate objective. Indeed, it became the key to formulating the Information Theory of Aging.

Now that the Werner gene, known as WRN, had been identified in humans, the next step was to test if the similar gene in yeast had the same function. If so, we could use yeast to more rapidly determine the cause of Werner syndrome and perhaps help us better understand aging in general. I marched into Guarente's office to tell him I was now studying Werner's syndrome in yeast and that's how we would solve aging.

In yeast, the equivalent of the WRN gene is Slow Growth Suppressor 1, or *SGS1*. The gene was already suspected to code for a type of enzyme called a

DNA helicase that untangles tangled strands of DNA before they break. Helicases are especially important in repetitive DNA sequences that are inherently prone to tangling and breaking. Functionality of proteins, such as the ones coded for by the Werner gene, is therefore vital, since more than half of our genome is, in fact, repetitive.

Through a gene-swapping process in which cells are tricked into picking up extra pieces of DNA, we swapped out the functional *SGS1* gene with a mutant version. In effect, we were testing to see if it was possible to give the yeast Werner syndrome.

After the swap, the yeast cells' lifespan was cut in half. Ordinarily, this would not have been news. Many events unrelated to aging—such as being eaten by a mite, drying out on a grape, or being placed in an oven—can and do shorten the lifespan of yeast cells. And here we'd messed with their DNA, which could have short-circuited the cells in a thousand different ways to cause early death.

But those cells weren't *just* dying. They were dying after a precipitous decline in health and function. As the *SGS1* mutants became older, they slowed down in their cell cycle. They grew larger. Both male and female “mating-type” genes (descendants of gene A) were switched on at the same time, so they were sterile and couldn't mate. These were all known hallmarks of aging in yeast. And it was happening more quickly in the mutants we'd made. It certainly looked like a yeast version of Werner's.

Using specialized stains, we colored the DNA blue and used red for the nucleolus, which sits inside the nucleus of all eukaryotic cells. That made it easier to see under the microscope what was happening at a cellular level.

And what was happening was fascinating.

The nucleolus is a part of the nucleus in which ribosomal DNA, or rDNA, resides. rDNA is copied into ribosomal RNA, which is used by ribosome enzymes to stitch amino acids together to make every new protein.

In the aged *SGS1* cells, the nucleolus looked as if it had exploded. Instead of a single red crescent swimming in a blue ocean, the nucleolus was scattered into half a dozen small islands. It was tragic and beautiful. The picture, which would later appear in the August 1997 issue of the prestigious journal *Science*, still hangs in my office.

What happened next was both enchanting and illuminating. In response to the damage, like rats to the call of the Pied Piper, the protein called Sir2—the first known sirtuin, which is encoded by the gene *SIR2*¹⁰ and descended from gene B—had moved away from the mating genes that control fertility and into the nucleolus.

That was a beautiful sight to me, but it was a problem for the yeast. Sir2 has an important job: it is an epigenetic factor, an enzyme that sits on genes, bundles up the DNA, and keeps them silent. At the molecular level, Sir2 achieves this via its enzymatic activity, making sure that chemicals called acetyls don't accumulate on the histones and loosen the DNA packaging.

When sirtuins left the mating genes—the ones descended from gene A that controlled fertility and reproduction—the mutant cells turned on both male and female genes, causing them to lose their sexual identity, just as in normal old cells, but much earlier.

I didn't understand at first why the nucleolus was exploding, let alone why the sirtuins were moving toward it as the cells grew older. I agonized over the question for weeks.

And then one night, after a long day in the lab, I woke up with an idea.

It came in the space between sleep-deprived delirium and deep dreaming. The wisps of a concept. A few words jumbled together. A muddled picture of something. That was enough, though, to jolt me awake and pull me from my bed.

I grabbed my notebook and went to the kitchen. There, hunched over the table in the early morning hours of October 28, 1996, I began to write.

*A theory on replicative senescence in yeast &
other organisms. -*

I wrote for about an hour, jotting down ideas, drawing pictures, sketching out graphs, formulating new equations.¹¹ Scientific observations that had previously made no sense to me were falling perfectly into a larger picture. Broken DNA causes genome instability, I wrote, which distracts the Sir2 protein, which changes the epigenome, causing the cells to lose their identity

and become sterile while they fixed the damage. Those were the analog scratches on the digital DVDs. Epigenetic changes cause aging.

There was, I imagined, a singular process that controlled them all. Not a countless number of separate cellular changes or diseases. Not even a set of hallmarks that could be addressed one at a time. There was something bigger—and more singular—than any of that.

This was the foundation for understanding the survival circuit and its role in aging.

The next day I showed Guarente my notes. I was excited; it felt like the biggest idea I'd ever had. But I was nervous, too; afraid he would find a hole in my logic and tear it apart. Instead, he looked over my notebook quietly, asked a few questions, and sent me on my way with six words.

“I like it,” he said. “Go prove it.”

THE RECITAL

To understand the Information Theory of Aging, we need to pay another visit to the epigenome, the part of the cell that the sirtuins help control.

Up close, the epigenome is more complex and wonderful than anything we humans have invented. It consists of strands of DNA wrapped around spooling proteins called histones, which are bound up into bigger loops called chromatin, which are bound up into even bigger loops called chromosomes.

Sirtuins instruct the histone spooling proteins to bind up DNA tightly, while they leave other regions to flail around. In this way, some genes stay silent, while others can be accessed by DNA-binding transcription factors that turn genes on.¹² Accessible genes are said to be in “euchromatin,” while silent genes are in “heterochromatin.” By removing chemical tags on histones, sirtuins help prevent transcription factors from binding to genes, converting euchromatin into heterochromatin.

Every one of our cells has the same DNA, of course, so what differentiates a nerve cell from a skin cell is the epigenome, the collective term for the control systems and cellular structures that tell the cell which genes should be turned on

and which should remain off. And this, far more than our genes, is what actually controls much of our lives.

One of the best ways to visualize this is to think of our genome as a grand piano.¹³ Each gene is a key. Each key produces a note. And from instrument to instrument, depending on the maker, the materials, and the circumstances of manufacturing, each will sound a bit different, even if played the exact same way. These are our genes. We have about 20,000 of them, give or take a few thousand.¹⁴

Each key can also be played *pianissimo* (soft) or *forte* (with force). The notes can be *tenuto* (held) or *allegretto* (played quickly). For master pianists, there are hundreds of ways to play each individual key and endless ways to play the keys together, in chords and combinations that create music we know as jazz, ragtime, rock, reggae, waltzes, whatever.

The pianist that makes this happen is the epigenome. Through a process of revealing our DNA or bundling it up in tight protein packages, and by marking genes with chemical tags called methyls and acetyls composed of carbon, oxygen, and hydrogen, the epigenome uses our genome to make the music of our lives.

Yes, sometimes the size, shape, and condition of a piano dictate what a pianist can do with it. It's tough to play a concerto on an eighteen-key toy piano, and it's mighty hard to make beautiful music on an instrument that hasn't been tuned in fifty years. Likewise, the genome certainly dictates what the epigenome can do. A caterpillar can't become a human being, but it can become a butterfly by virtue of changes in epigenetic expression that occur during metamorphosis, even though its genome never changes. Similarly, the child of two parents from a long line of people with black hair and brown eyes isn't likely to develop blond hair and blue eyes, but twin agouti mice in the lab can turn out brown or golden, depending on how much the *Agouti* gene is turned on during gestation by environmental influences on the epigenome, such as folic acid, vitamin B₁₂, genistein from soy, or the toxin bisphenol A.¹⁵

Similarly, among monozygotic human twins, epigenetic forces can drive two people with the same genome in vastly different directions. It can even cause them to age differently. You can see this clearly in side-by-side photographs of the faces of smoking and nonsmoking twins; their DNA is still largely the same, but

the smokers have bigger bags under their eyes, deeper jowls below their chins, and more wrinkles around their eyes and mouths. They are not older, but they've clearly aged faster. Studies of identical twins place the genetic influences on longevity at between 10 and 25 percent which, by any estimation, is surprisingly low.¹⁶

Our DNA is not our destiny.

Now imagine you're in a concert hall. A virtuoso pianist is seated at a gorgeously polished Steinway grand. The concerto begins. The music is beautiful, breathtaking. Everything is perfect.

But then, a few minutes into the piece, the pianist misses a key. The first time it happens, it's almost unnoticeable—an extra D, perhaps, in a chord that doesn't need that note. Embedded in so many perfectly played notes, hidden among an otherwise flawless chord in an otherwise perfect melody, it's nothing to worry about. But then, a few minutes later, it happens again. And then, with increasing frequency, again and again and again.

It's important to remember that there is nothing wrong with the piano. And the pianist is playing most of the notes prescribed by the composer. She's just also playing some extra notes. Initially, this is just annoying. Over time it becomes unsettling. Eventually it ruins the concerto. Indeed, we'd assume that there was something wrong with the pianist. Someone might even rush onto the stage to make sure she is all right.

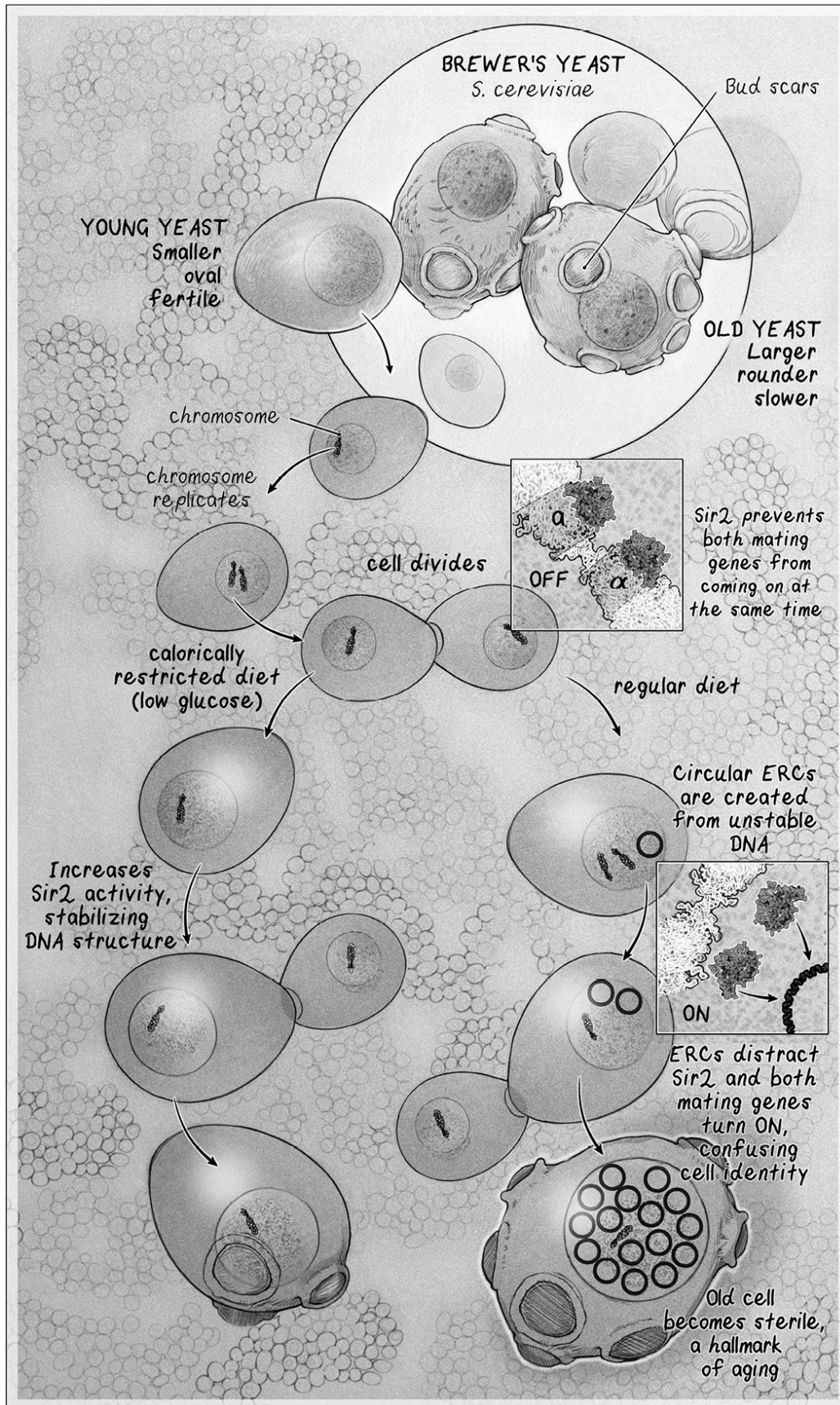
Epigenetic noise causes the same kind of chaos. It is driven in large part by highly disruptive insults to the cell, such as broken DNA, as it was in the original survival circuit of *M. superstes* and in the old yeast cells that lost their fertility. And this, according to the Information Theory of Aging, is why we age. It's why our hair grays. It's why our skin wrinkles. It's why our joints begin to ache. Moreover, it's why each one of the hallmarks of aging occurs, from stem cell exhaustion and cellular senescence to mitochondrial dysfunction and rapid telomere shortening.

This is, I acknowledge, a bold theory. And the strength of a theory is based on how well it predicts the results of rigorous experiments, often millions of them, the number of phenomena it can explain, and its simplicity. The theory was

simple, and it explained a lot. As good scientists, what we had left to do was to try our best to disprove it and see how long it survived.

To get started, Guarente and I had to get our eyes on some yeast DNA.

We used a technique called a Southern blot, a method of separating DNA based on its size and conformation and lighting it up with a radioactive DNA probe. In the first experiment, we noticed something spectacular. Normally, the rDNA of a yeast cell that is made visible by a Southern blot is tightly packed, like a new spool of rope, with a few barely visible wispy loops of supercoiled DNA. But the rDNA of the yeast cells we'd created in our lab—the Werner mutants that seemed to be aging rapidly—were madly unpacking, like a vacuum-sealed bag of yarn that had been ripped open.



LESSONS FROM YEAST CELLS ABOUT WHY WE AGE. In young yeast cells, male and female “mating-type information” (gene A) is kept in the “off” position by the Sir2 enzyme, the first sirtuin (encoded by a descendant of gene B). The highly repetitive ribosomal DNA (rDNA) is unstable, and toxic DNA circles form; these recombine and eventually accumulate to toxic levels in old cells, killing them. In response to DNA circles and the perceived genome instability, Sir2 moves away from silent mating-type genes to help stabilize the genome. Both male and female genes turn on, causing infertility, the main hallmark of yeast aging.

The rDNA was in a state of chaos. The genome, it seemed, was fragmenting. DNA was recombining and amplifying, showing up on the Southern blot as dark spots and wispy circles, depending on how coiled up and twisted they were. We called those loops extrachromosomal ribosomal DNA circles, or ERCs, and they were accumulating as the mutants aged.

If we had indeed induced aging, then we would see this same pattern emerge in yeast cells that had aged normally.

We don’t count the age of a single yeast cell with birthday candles. They simply don’t last that long. Instead, aging in yeast is measured by the number of times a mother cell divides to produce daughter cells. In most cases, a yeast cell gets to about 25 divisions before it dies. That, however, makes obtaining old yeast cells an exceptionally challenging task. Because by the time an average yeast cell expires, it is surrounded by 2^{25} , or 33 million, of its descendants.

It took a week of work, a lot of sleepless nights, and a whole lot of caffeinated beverages to collect enough regular old cells. The next day, when I developed the film to visualize the rDNA, what I saw astounded me.¹⁷

Just like the mutants, the normal old yeast cells were packed with ERCs.

That was a “Eureka!” moment. Not proof—a good scientist never has proof of anything—but the first substantial confirmation of a theory, the foundation upon which I and others would build more discoveries in the years to come.

The first testable prediction was if we put an ERC into very young yeast cells—and we devised a genetic trick to do that—the ERCs would multiply and distract the sirtuins, and the yeast cells would age prematurely, go sterile, and die young—and they did. We published that work in December 1997 in the scientific journal *Cell*, and the news broke around the world: “Scientists figured out a cause of aging.”

It was there and then that Matt Kaeberlein, a PhD student at the time, arrived at the lab. His first experiment was to insert an extra copy of *SIR2* into the genome of yeast cells to see if it could stabilize the yeast genome and delay aging. When the extra *SIR2* was added, ERCs were prevented, and he saw a 30 percent increase in the yeast cells' lifespan, as we'd been hoping. Our hypothesis seemed to be standing up to scrutiny: the fundamental, upstream cause of sterility and aging in yeast was the inherent instability of the genome.

What emerged from those initial results in yeast, and another decade of pondering and probing mammalian cells, was a completely new way to understand aging, an information theory that would reconcile seemingly disparate factors of aging into one universal model of life and death. It looked like this:

Youth → broken DNA → genome instability → disruption of DNA packaging and gene regulation (the epigenome) → loss of cell identity → cellular senescence → disease → death.

The implications were profound: if we could intervene in any of these steps, we might help people live longer.

But what if we could intervene in all of them? Could we stop aging?

Theories must be tested and tested and tested some more—not just by one scientist but by many. And to that end, I was fortunate to have been put onto a research team that included some of the most brilliant and insightful scientists in the world.

There was Lenny Guarente, our indefatigable mentor. There was also Brian Kennedy, who started the yeast-aging project in Lenny's lab and has since played a tremendously important role in understanding premature aging diseases and the impact of genes and molecules that increase health and longevity in model organisms. There were Monica Gotta and Susan Gasser at the University of Geneva, who are now some of the most influential researchers in the field of gene regulation; Shin-ichiro Imai, now a professor at Washington University, who discovered that sirtuins are NAD-utilizing enzymes and now does research into how the body controls sirtuins; Kevin Mills, who ran a lab in Maine, then

became a cofounder of and chief scientific officer at Cyteir Therapeutics, which develops novel ways to fight cancer and autoimmune diseases; Nicanor Austriaco, who started the project with Brian, now teacher of biology and theology at Providence College, a great combo; Tod Smeal, chief scientific officer of cancer biology at the global pharmaceutical company Eli Lilly; David Lombard, who is now a researcher in the field of aging at the University of Michigan; Matt Kaeberlein, a professor at the University of Washington, who is testing molecules on dog longevity; David McNabb, whose lab at the University of Arkansas has made key and lifesaving discoveries about fungal pathogens; Bradley Johnson, an expert on human aging and cancer at the University of Pennsylvania; and Mala Murthy, a prominent neuroscientist now at Princeton.

Again and again I have been greatly privileged in the matter of those who work around me. And that was never truer than it was in Guarente's lab at MIT. It was a dream team, and I often felt humbled by the people with whom I was surrounded.

When I began my career in this field, I dreamt of publishing just one study in a top-tier journal. During those years, our group was publishing one every few months.

We demonstrated that the redistribution of Sir2 to the nucleolus is a response to numerous DNA breakages, which happen as a result of ERCs multiplying and inserting back into the genome or joining together to form superlarge ERCs. When Sir2 moves to combat DNA instability, it causes sterility in old, bloated yeast cells. That was the first step of the survival circuit, though at the time we had no idea that it was as ancient and as essential to our very existence as it turned out to be.

We told the world that we could give yeast a Werner-like syndrome, causing exploded nucleoli.¹⁸ We described the ways in which mutants of *SGS1*, those we'd plagued with the yeast equivalent to the Werner syndrome mutation, accumulated ERCs more rapidly, leading to premature aging and a shortened lifespan.¹⁹ Crucially, by demonstrating that if you add an ERC to young cells they age prematurely, we had crucial evidence that ERCs don't just happen during aging, they cause it. And by artificially breaking the DNA in the cell and watching the cellular response, we showed why sirtuins move—to help with

DNA repair.²⁰ That turned out to be the second step of the survival circuit.²¹ The DNA damage that gave rise to the ERCs was distracting Sir2 from the mating-type genes, causing them to become sterile, a hallmark of yeast aging.

It was epigenomic noise in its purest form.

It took another twenty years to learn if those findings in yeast were relevant to organisms more complex than yeast. We mammals have seven sirtuin genes that have evolved a variety of functions beyond what simple *SIR2* can do. Three of them, *SIRT1*, *SIRT6*, and *SIRT7*, are critical to the control of the epigenome and DNA repair. The others, *SIRT3*, *SIRT4*, and *SIRT5*, reside in mitochondria, where they control energy metabolism, while *SIRT2* buzzes around the cytoplasm, where it controls cell division and healthy egg production.

There had been many clues along the way. Brown University's Stephen Helfand showed that adding extra copies of the dSir2 gene to fruit flies suppresses epigenetic noise and extends their lifespan. We found that SIRT1 in mammals moves from silent genes to help repair broken DNA in mouse and human cells.²² But the true extent to which the survival circuit is conserved between yeast and humans wasn't fully known until 2017, when Eva Bober's team at the Max Planck Institute for Heart and Lung Research in Bad Nauheim, Germany, reported that sirtuins stabilize human rDNA.²³ Then, in 2018, Katrin Chua at Stanford University found that, by stabilizing human rDNA, sirtuins prevent cellular senescence—essentially the same antiaging function as we had found for sirtuins in yeast twenty years earlier.²⁴

That was an astonishing revelation: over a billion years of separation between yeast and us, and, in essence, the circuit hadn't changed.

By the time those findings appeared, though, it was clear to me that epigenomic noise was a likely catalyst of human aging. Two decades of research had already been leading us in that direction.²⁵

In 1999, I moved from MIT across the river to Harvard Medical School, where I set up a new lab on aging. There I was hoping to answer a new question that had increasingly been occupying my thoughts.

I had noticed that yeast cells fed with lower amounts of sugar were not just living longer, but their rDNA was exceptionally compact—significantly delaying

the inevitable ERC accumulation, catastrophic numbers of DNA breaks, nucleolar explosion, sterility, and death.

Why was that happening?

THE SURVIVAL CIRCUIT COMES OF AGE

Our DNA is constantly under attack. On average, each of our forty-six chromosomes is broken in some way every time a cell copies its DNA, amounting to more than 2 trillion breaks in our bodies per day. And that's just the breaks that occur during replication. Others are caused by natural radiation, chemicals in our environment, and the X-rays and CT scans that we're subjected to.

If we didn't have a way to repair our DNA, we wouldn't last long. That's why, way back in primordium, the ancestors of every living thing on this planet today evolved to sense DNA damage, slow cellular growth, and divert energy to DNA repair until it was fixed—what I call the survival circuit.

Since the yeast work, evidence that yeast aren't so different from us has continued to accumulate. In 2003, Michael McBurney from the University of Ottawa in Canada discovered that mouse embryos manipulated to be unable to produce one of the seven sirtuin enzymes, SIRT1, couldn't last past the fourteenth day of development—about two-thirds of the way into a mouse's gestation period.²⁶ Among the reasons, the team reported in the journal *Cancer Cell*, was an impaired ability to respond to and repair DNA damage.²⁷ In 2006, Frederick Alt, Katrin Chua, and Raul Mostoslavsky at Harvard showed that mice engineered to lack SIRT6 underwent the typical signs of aging faster along with shortened lifespans.²⁸ When the scientists knocked out a cell's ability to create this vital protein, the cell lost its ability to repair double-strand DNA breaks, just as we had showed in yeast back in 1999.

If you are skeptical, and you should be, you might assume these SIRT mutant mice could just be sick and, therefore, short lived. But adding in more copies of the sirtuin genes *SIRT1* and *SIRT6* does just the opposite: it increases the health and extends the lifespan of mice, just as adding extra copies of the yeast *SIR2*

gene does in yeast.²⁹ Credit for these discoveries goes to two of my previous colleagues, Shin-ichiro Imai, my former drinking buddy at the Guarente lab, and Haim Cohen, my first postdoc at Harvard.

In yeast, we had shown that DNA breaks cause sirtuins to relocate away from silent mating-type genes, causing old cells to become sterile. That was a simple system, and we'd figured it out in a few years.

But is the survival circuit causing aging in mammals? What parts of the system survived the billion years, and which are yeast specific? Those questions are on the cutting edge of human knowledge right now, but the answers are beginning to reveal themselves.

What I'm suggesting is that the *SIR2* gene in yeast and the *SIRT* genes in mammals are all descendants of gene B, the original gene silencer in *M. superstes*.³⁰ Its original job was to silence a gene that controlled reproduction.

In mammals, the sirtuins have since taken on a variety of new roles, not just as controllers of fertility (which they still are). They remove acetyls from hundreds of proteins in the cell: histones, yes, but also proteins that control cell division, cell survival, DNA repair, inflammation, glucose metabolism, mitochondria, and many other functions.

I've come to think of sirtuins as the directors of a multifaceted disaster response corps, sending out a variety of specialized emergency teams to address DNA stability, DNA repair, cell survivability, metabolism, and cell-to-cell communication. In a way, this is like the command center for the thousands of utility workers who descended upon Louisiana and Mississippi in the wake of Hurricane Katrina in 2005. Most of the workers weren't from the Gulf Coast, but they came, did their level best to fix what was broken, and then went home. Some were working in the storm-ravaged communities for a few days and others for a few weeks before returning to their normal lives. And for most, it wasn't the first or last time they had done something like that; anytime there's a mass disaster that impacts utilities, they swoop in to help.

When they're home, those folks take care of the typical business of being at home: paying bills, mowing lawns, coaching baseball, whatever. But when they're away, helping keep places like the Gulf Coast from descending into

anarchy—a condition that would have had disastrous results for the rest of the nation—a lot of those things have to be put on hold.

When sirtuins shift from their typical priorities to engage in DNA repair, their epigenetic function at home ends for a bit. Then, when the damage is fixed and they head back to home base, they get back to doing what they usually do: controlling genes and making sure the cell retains its identity and optimal function.

But what happens when there's one emergency after another to tend to? Hurricane after hurricane? Earthquake after earthquake? The repair crews are away from home a lot. The work they normally do piles up. The bills come due, then overdue, and then the folks from collections start calling. The grass grows too long, and soon the president of the neighborhood association is sending nastygrams. The baseball team goes coachless, and the team devolves into the Bad News Bears. And most of all, one of the most important things they do while at home—reproducing—doesn't get done. This form of hormesis, the original survival circuit, works fine to keep organisms alive in the short term. But unlike longevity molecules that simply mimic hormesis by tweaking sirtuins, mTOR, or AMPK, sending out the troops on fake emergencies, these real emergencies create life-threatening damage.

What could cause so many emergencies? DNA damage. And what causes that? Well, over time, *life* does. Malign chemicals. Radiation. Even normal DNA copying. These are the things that we've come to believe are the causes of aging, but there is a subtle but vital shift we have to make in that manner of thinking. It's not so much that the sirtuins are overwhelmed, though they probably are when you are sunburned or get an X-ray; what's happening every day is that the sirtuins and their coworkers that control the epigenome don't always find their way back to their original gene stations after they are called away. It's as if a few emergency workers who came to address the damage done in the Gulf Coast by Katrina had lost their home address. Then disaster strikes again and again, and they must redeploy.

Wherever epigenetic factors leave the genome to address damage, genes that should be off, switch on and vice versa. *Wherever* they stop on the genome, they

do the same, altering the epigenome in ways that were never intended when we were born.

Cells lose their identity and malfunction. Chaos ensues. The chaos materializes as aging. This is the epigenetic noise that is at the heart of our unified theory.

How does the *SIR2* gene actually turn off genes? *SIR2* codes for a specialized protein called a histone deacetylase, or HDAC, that enzymatically cleaves the acetyl chemical tags from histones, which, as you'll recall, causes the DNA to bundle up, preventing it from being transcribed into RNA.

When the Sir2 enzyme is sitting on the mating-type genes, they remain silent and the cell continues to mate and reproduce. But when a DNA break occurs, Sir2 is recruited to the break to remove the acetyl tags from the histones at the DNA break. This bundles up the histones to prevent the frayed DNA from being chewed back and to help recruit other repair proteins. Once the DNA repair is complete, most of the Sir2 protein goes back to the mating-type genes to silence them and restore fertility. That is, unless there is another emergency, such as the massive genome instability that occurs when ERCs accumulate in the nucleoli of old yeast cells.

For the survival circuit to work and for it to cause aging, Sir2 and other epigenetic regulators must occur in "limiting amounts." In other words, the cell doesn't make enough Sir2 protein to simultaneously silence the mating-type genes *and* repair broken DNA; it has to shuttle Sir2 between the various places on an "as-needed" basis. This is why adding an extra copy of the *SIR2* gene extends lifespan and delays infertility: cells have enough Sir2 to repair DNA breaks *and* enough Sir2 to silence the mating-type genes.³¹

Over the past billion years, presumably millions of yeast cells have spontaneously mutated to make more Sir2, but they died out because they had no advantage over other yeast cells. Living for 28 divisions was no advantage over those that lived for 24 and, because Sir2 uses up energy, having more of the protein may have even been a disadvantage. In the lab, however, we don't notice any disadvantage because the yeast are given more sugar than they could possibly ever eat. By adding extra copies of the *SIR2* gene, we gave the yeast cells what evolution failed to provide.